

Yuwei Liu

410-500-7576 | yliu689@jh.edu | linkedin.com/in/yuwei-liu-137338380 | yuwei-liu01.github.io

RESEARCH INTERESTS

Surgical robotics, medical robotics, and robot learning, with an emphasis on connecting perception, planning, and control for intelligent and autonomous robotic assistance.

EDUCATION

Johns Hopkins University Aug. 2025 – May 2027
Master of Science in Mechanical Engineering, Robotics Track Baltimore, MD
• Coursework: Computer Integrated Surgery, Machine Learning, Robot Motion Planning, Robot Kinematics & Dynamics, Control System Design, Haptics & HRI, Algorithms for Sensor-Based Robotics
Wuhan University of Science and Technology (WUST) Sep. 2016 – June 2021
Bachelor of Engineering in Architecture Wuhan, China
• Rank: Top 10% | GPA: 3.45/4.0 | Outstanding Graduate | Academic Excellence Scholarship, 3 consecutive years

RESEARCH EXPERIENCE

Research Assistant, Learning-Based Robotic Cholecystectomy Feb. 2026 – Present
ARCADE Lab, Johns Hopkins University Baltimore, MD
• Contribute to a clinically informed surgical robotics project for autonomous robotic cholecystectomy, developing perception and data infrastructure to support learning-based manipulation.
• Built and evaluated an automated YOLO + SAM3 surgical video segmentation workflow to generate anatomy and instrument masks across 5,000+ ex-vivo robotic surgery episodes, with warning logs and review outputs to flag technical failures.
• Investigated a lightweight spatial CNN refiner for inter-class mask bleeding and developed class-balanced metrics and pairwise confusion analyses for evaluation on imbalanced anatomical structures.
• Reproduced and instrumented a learning-based surgical manipulation pipeline to analyze failure modes in model inference, autonomous rollouts, and perception-action integration.

PROJECTS

Haptic Feedback Gripper for Robotic Surgery Training | *C++, Arduino, SOFA, SolidWorks* Sep. 2025 – Dec. 2025
• **1st Place Course Project:** Led a 4-person team to build a real-time haptic feedback gripper for robotic surgery training, integrating mechanical design, motor control, and tissue simulation.
• Coupled a direct-drive BLDC motor with SOFA simulation in a hardware-in-the-loop system for force-feedback interaction.
• Implemented a two-loop control strategy bridging 200 Hz simulation and 1000 Hz motor control, improving stability during contact-force rendering.
UR5 Robot Kinematics, Motion Planning, and Control | *ROS 2, MATLAB, Python* Feb. 2026 – Apr. 2026
• Implemented Product-of-Exponentials forward/inverse kinematics for a UR5 robotic arm and validated end-effector trajectories against planned Cartesian paths.
• Built a ROS 2 motion execution workflow for waypoint tracking, trajectory interpolation, and joint-space movement.
• Improved motion stability near singular configurations through planning-parameter tuning and trajectory validation.
Multi-Robot Collaborative Construction | *Python, Arduino, LiDAR, Vision* Oct. 2024 – Nov. 2024
• Designed a 3-robot construction workflow to assemble a 5x5 reciprocal shell using coordinated sensing, localization, and construction sequencing
• Integrated LiDAR and vision-based localization signals to support real-time positioning, evaluate construction accuracy, and reduce assembly errors

SELECTED PUBLICATION

[Third Prize] Y. Liu and F. Wang, “A Study on the Architecture of a Nursing Home Alongside Kindergarten: The Elderly Apartment in Tongxin Garden,” *Hubei Civil Engineering Academy Journal*, no. 4, 2019.

PROFESSIONAL EXPERIENCE

Assistant Architect, Urban Infrastructure Division June 2021 – June 2024
Central South Architectural Design Institute (CSADI) Wuhan, China
• Applied parametric modeling and BIM workflows to resolve geometric constraints, optimize design feasibility, and coordinate engineering requirements across disciplines
• Led digital workflow training for 15 team members on computational design tools, improving team efficiency and design quality

TECHNICAL SKILLS

Programming & Tools: Python, C++, MATLAB/Simulink, Bash, Git, Linux/Ubuntu, Arduino
Robotics: ROS 2, kinematics, motion planning, trajectory generation, haptic control, perception pipelines
Design: SolidWorks, AutoCAD, Rhino, Grasshopper, BIM, 3D printing